

Proof Of Concept SRS

1. Introduction

1.1 Purpose

The purpose of Carbon Credit Exchange trading platform is to provide a digital platform that converts verified environmentally sustainable activities into tradable carbon credits while ensuring automation, fairness, and transparency throughout trading.

1.2 Scope

The system operates as a secure and transparent marketplace connecting Sellers and Buyers through the generation, verification, trading, and management of carbon credits derived from verified sustainability activities.

1.3 Intended Users

The paper explicitly identifies the following user groups.

User
Seller
Buyer
Admin

2. Overall Description

2.1 Product Functions

Function
Sustainability action submission
Carbon credit calculation
Credit verification
Credit activation
Credit listing
Credit purchase

Credit transfer
Transaction logging

2.2 Dependencies

The paper explicitly states the following external or operational dependencies.

Dependency
Verified sustainability evidence
Standard environmental metrics
Minimum market price defined by the platform(admin)

3. Functional Requirements

ID	Requirement
FR-B-01	The system shall allow Buyers to register into an account.
FR-B-02	The system shall allow Buyers to log in.
FR-B-03	The system shall allow Buyers to manage their accounts securely.
FR-B-04	The system shall allow Buyers to browse available verified carbon credits.
FR-B-05	The system shall rank available credit listings using a weighted score based on validity, price, and quantity.
FR-B-06	The system shall allow Buyers to purchase carbon credits.
FR-B-07	The system shall perform compliance checks during every purchase.
FR-B-08	The system shall validate buyer eligibility during every purchase.
FR-B-09	The system shall validate purchase limits during every purchase.
FR-B-10	The system shall validate rule conformance as defined by the Admin during every purchase.
FR-B-11	The system shall automatically generate a digital receipt after a successful purchase.
FR-B-12	The system shall include a unique ID in the digital receipt.
FR-B-13	The system shall include transaction details in the digital receipt.

FR-B-14	The system shall allow Buyers to view their purchased carbon credits.
FR-B-15	The system shall allow Buyers to monitor the balance of their purchased credits.
FR-B-16	The system shall allow Buyers to monitor the expiry status of their purchased credits.
FR-A-01	The system shall authenticate Admin users securely.
FR-A-02	The system shall allow Admin users to manage user access roles.
FR-A-03	The system shall allow Admin users to review submitted seller actions.
FR-A-04	The system shall allow Admin users to approve submitted seller actions based on the provided evidence.
FR-A-05	The system shall allow Admin users to reject submitted seller actions based on the provided evidence.
FR-A-06	The system shall issue verified carbon credits upon approval.
FR-A-07	The system shall assign a unique UUID to verified carbon credits upon approval.
FR-A-08	The system shall assign an expiry period to verified carbon credits upon approval.
FR-A-09	The system shall allow Admin users to define the minimum selling price per credit.
FR-A-10	The system shall allow Admin users to update the minimum selling price per credit.
FR-A-11	The system shall allow Admin users to monitor trading data.
FR-A-12	The system shall allow Admin users to monitor transactions.
FR-A-13	The system shall allow Admin users to monitor system health.
FR-A-14	The system shall allow Admin users to configure automatic expiry alerts.
FR-A-15	The system shall allow Admin users to export analytical reports summarising trades, verifications, and market statistics.
FR-A-16	The system shall allow Admin users to approve user accounts.
FR-A-17	The system shall allow Admin users to suspend user accounts.
FR-A-18	The system shall allow Admin users to delete user accounts.
FR-UC-01	The system shall roll back the transaction if payment or compliance validation fails during credit purchase.
FR-UC-02	The system shall reserve credits after processing payment.
FR-UC-03	The system shall update the buyer's portfolio after a successful purchase.

FR-UC-04	The system shall decrease the seller's stock after a successful purchase.
FR-UC-05	The system shall display a confirmation message after a successful purchase.
FR-UC-06	The system shall prompt the buyer to select another listing when credits are unavailable.
FR-UC-07	The system shall discard the transaction if the buyer cancels before confirmation.
FR-UC-08	The system shall notify the buyer when payment fails.
FR-UC-09	The system shall retry receipt generation when a receipt generation error occurs.
FR-UC-10	The system shall save a listing as a draft when a network or database error occurs during listing.
FR-UC-11	The system shall reject listings priced below the minimum threshold.
FR-UC-12	The system shall prompt the seller to revise listings priced below the minimum threshold.
FR-UC-13	The system shall allow sellers to retry listing submission after a network or database error.
FR-UC-14	The system shall keep a listing visible but unranked when broker ranking fails until recalculated.

4. Non-functional Requirements

Only **explicitly stated** non-functional requirements have been extracted from the paper. No quality attribute has been inferred.

ID	Category	Requirement
NFR-01	Extensibility	The system shall be easy to extend in future releases without major structural changes.
NFR-02	Extensibility	The system shall allow new actor roles to be added with minimal code refactoring.
NFR-03	Extensibility	The system shall allow rule definitions to be added via configuration rather than hard-coded logic.
NFR-04	Security	The system shall secure all communication using HTTPS and JWT-based authentication (JSON Web Token).
NFR-05	Reliability / Auditability	The system shall generate a verifiable UUID for each transaction and log it for auditability.
NFR-06	Reliability	The system shall prevent race conditions during verification and trading operations through transaction-safe methods.
NFR-07	Performance	The purchase transaction shall complete within one minute.
NFR-08	Performance	Credit listings shall be published within one minute of submission.

5. Constraints

ID	Constraint
CON-01	Sellers must accumulate a minimum of 10 carbon credits before submitting them for Admin verification.
CON-02	Carbon credits begin to depreciate after 70% of their validity period.
CON-03	Sellers shall not set a selling price below the minimum market price established by the platform.
CON-04	Verification requests must include supporting documentation.
CON-05	Only verified and approved carbon credits become active and tradable.

6. Assumptions

ID	Assumption
A-01	Buyer account exists and is logged in before purchasing credits.
A-02	Credits listed for purchase are verified and available.
A-03	The Seller has successfully logged in before listing credits.
A-04	The Seller already possesses verified carbon credits issued by the Admin.
A-05	Minimum price and expiry policies are predefined in the system.